

Vision Document

HPCRMS High Priority Car Rental Management System

Team members:

Orion Blackstock 618684

1. Vision Statement

The vision of the High Priority Car Rental Management System is to transform traditional vehicle rental operations into a seamless digital experience by minimizing customer wait times, automating rental processes, and providing fast, secure, and convenient access to vehicles. The system will improve operational efficiency, maximize fleet utilization, and deliver superior customer experience across all rental locations.

1.1 Introduction

Traditional car rental companies often rely on manual processes for customer registration, identity verification, vehicle allocation, contract preparation, and payment processing. During peak travel seasons and holidays, these processes result in long queues, delayed vehicle pickups, frustrated customers, and inefficient use of staff resources.

Customers expect quick, digital, and self-service experiences similar to airline check-ins and ride-sharing applications. However, many rental agencies still require customers to complete paperwork at the rental counter even after making online reservations.

The **High Priority Car Rental Management System (HPCRMS)** is designed to digitize and automate the vehicle rental process, enabling customers to reserve, verify their identity, complete documentation, and receive vehicle access with minimal waiting time. The system will improve operational efficiency while enhancing customer satisfaction.

2. Positioning

2.1 Problem Statement

The problem of	Long wait time for customers and inefficient car rental processing
Affects	Customer, Rental Agents, Branch Managers, Fleet Managers
the impact of which is	Long line at the counter, slow rental pick up process, customer dissatisfaction, loss of business opportunities due to long line, increased rental cancellations and poor fleet utilization
a successful solution would be	A centralized Car Rental Management System that automates reservations, customer

	verification, digital contracts, vehicle allocation, and payments while providing customers with fast and convenient vehicle access through web and mobile platforms.
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2.2 Product Position Statement

For	Car rental companies are seeking to improve customer experience and operational efficiency.
Who	Need to reduce customer waiting times, streamline rental operations, and improve fleet utilization.
High Priority Car Rental Management System (HPCRMS)	is a game changer in the car rental industry this solution offers a web and mobile platform for fleet management and vehicle rental solution.
That	Automates the entire rental lifecycle from reservation to vehicle return, allowing customers to receive vehicles quickly with minimal staff intervention.
Unlike	Current rental process that allows you to reserve cars online still requires you to complete the process over the counter
Our product	Provides real-time vehicle availability, automated document processing, digital contracts, online payments, customer self-service, and intelligent vehicle assignment to significantly reduce waiting time.

3. Stakeholder Descriptions

3.1 Stakeholder Summary

Name	Description	Responsibilities
Customers	Reserve, rent, and return vehicles	Faster rentals, shorter wait times, convenient self-service
Rental Agents	Assist customers and manage rentals	Reduced paperwork and faster service
Branch Managers	Monitor branch operations	Improved operational efficiency and reporting
Fleet Managers	Manage vehicles and maintenance	Better fleet utilization and scheduling
Finance Department	Process payments and refunds	Accurate financial records and automated billing

System Administrators	Manage users and system configuration	Secure and reliable system management
Business and Quality Analyst	Use tools to produce documents, outlining requirements and creating test cases	BA's and QA's are responsible for gathering requirements, producing documentation and test system functionalities
Software Engineers	Engineering system based on requirements	Software Engineers are responsible for developing system features, maintaining application and package application for deployment

3.2 User Environment

The system will operate across multiple rental branches connected to a centralized cloud database.

Customers can access the system using:

- Mobile phones
- Tablets
- Desktop computers

Rental agents will use desktop computers or tablets to process rentals and assist customers when necessary.

The system will integrate with:

- Online payment gateways
- Driver's license verification services
- Vehicle GPS tracking systems
- Fleet maintenance systems
- Email and SMS notification services

The system will support multiple branches operating simultaneously while maintaining real-time vehicle availability.

4. Product Overview

4.1 Product Perspective

The Car Rental Management System will serve as the central platform for all rental operations. It will integrate customer management, reservation management, fleet management, payment processing, reporting, and maintenance scheduling into one unified system. Third party

application will receive secure API endpoints to access vehicle availability and payment processing.

Major components include:

- Customer Portal
- Mobile Compatibility
- Reservation Management
- Vehicle Inventory Management
- Fleet Maintenance
- Payment Processing
- Vehicle Pickup and Return
- Reporting Dashboard
- Administration Module

4.2 Assumptions and Dependencies

The system assumes:

- Internet connectivity is available at all rental branches.
- Customers possess valid driver's licenses.
- Vehicles are equipped with GPS tracking.
- Online payment gateways remain available.
- SMS and email services are operational.
- Vehicle availability is updated in real time.
- Branches share a centralized database.

4.3 Needs and Features

No.	Problem	Need	Priority	Feature	Planned Release
Customer Management					
1	Customer information is repeatedly entered for every rental.	Customer information should be stored securely for future rentals.	High	Registration, profile management, and rental history.	Reales
2	Staff manually verify driver's licenses.	Customer verification should be automated.		Automated validation of driver's license information.	Release
3	Customers are unaware of reservation updates.	Customers should receive automatic notifications.	Medium	Email and SMS confirmations, reminders, and updates.	Release

4	Customers spend time waiting to pick up reserved vehicles.	Customers need a faster pickup process.	High	Scan a QR code to retrieve reservation details and complete pickup.	Release
5	Rental companies struggle to retain repeat customers.	Customers should be rewarded for loyalty.		Reward points, discounts, and promotional offers.	
6	Customers spend time selecting suitable vehicles.	Customers need intelligent recommendations.		Recommend vehicles based on customer preferences and rental history.	
7	Customers experience long lines when making reservations.	Customers need a faster way to reserve vehicles before arriving at the branch.	High	Search vehicles, make, modify, and cancel reservations.	Release
8	Customers wait for manual payment processing.	Customers need secure online payment options.		Credit/debit card payments and digital receipts.	
Digital Check-In					
9	Customers spend excessive time completing paperwork.	Customers need a digital check-in process.	High	Upload driver's license, verify identity, and complete rental details online.	Release
10	Customers wait while rental agents prepare contracts.	Customers need electronic rental agreements.		Electronic signatures and automated contract generation.	
Rental and Vehicle Management					
11	Rental agents manually search for available vehicles.	Staff need real-time vehicle availability.		Real-time vehicle inventory and availability tracking.	
12	Vehicles are assigned manually,	Staff need automated vehicle		Assign the best available vehicle	

	causing delays.	assignment.		based on reservation criteria.	
13	Customers have limited visibility into available vehicles.	Customers need detailed vehicle information before booking.		Browse vehicle categories, specifications, pricing, and images.	
14	Rental agents manually record vehicle inspections.	Vehicle inspections should be digitized.	Medium	Record vehicle condition with photos and electronic forms.	Release
15	Vehicle maintenance schedules are difficult to manage.	Fleet managers need maintenance tracking.	Medium	Schedule servicing and monitor vehicle maintenance status.	Release
Reporting and Analytics					
16	Managers cannot easily monitor fleet utilization.	Managers need operational insights.	Medium	Reporting Dashboard – Rental trends, utilization reports, revenue, and branch performance.	Release 2
17	Management cannot accurately forecast vehicle demand.	Business needs predictive analytics.		Analytics Dashboard – Demand forecasting, fleet optimization, and customer behavior analysis.	
Security					
18	Administrators manage users manually.	Secure user administration is required.		Role-based access control for employees and administrators.	
19	Customer information must be protected.	The system requires strong security controls.		Authentication, authorization, encryption, and activity logs.	

4.4 Alternatives and Competition

Existing Manual Process

Strengths

- Minimal technology investment
- Familiar workflow

Weaknesses

- Long customer wait time
- High labor costs
- Manual errors
- Poor customer experience

TRAWEX

Strengths

- Established solutions
- Rich functionality

Weaknesses

- High licensing costs
- Limited customization
- Complex implementation

Proposed High Priority Car Rental Management System

Strengths

- *Reduced customer waiting time*
- *Automated rental workflow*
- *Customer self-service*
- *Real-time fleet visibility*
- *Improved customer satisfaction*
- *Scalable cloud architecture*
- *Lower operational costs*

Weaknesses

- *Initial implementation cost*
- *Staff training required*
- *Dependence on internet connectivity*

5. Other Product Requirements

Performance Requirements

- Vehicle search results displayed within 2 seconds.
- Reservation confirmation within 5 seconds.
- Support at least 1,000 concurrent users.
- 99.9% system availability.
- Real-time synchronization across all branches.

Usability Requirements

- User-friendly interface.
- Mobile-responsive design.

- Self-service workflow requires minimal training.
- Accessibility compliance.

Security Requirements

- Multi-factor authentication for employees.
- Encrypted customer information.
- Secure online payments.
- Compliance with data privacy regulations.
- Role-based authorization.

Platform Requirements

- Web application
- Android application
- iOS application
- Cloud-hosted database
- RESTful API architecture

Documentation Requirements

- User Manual
- Administrator Guide
- API Documentation
- Installation Guide
- Maintenance Guide
- Training Materials